

Class 1: Sentential Logic

Two goals for the course:

1.

2.

One type of logical argument called a syllogism consists of 3 statements.

1st two statements are called

The 3rd statement is called

Example 1

1. Every integer is a rational number.
2. Every rational number is a real number.
3. Therefore

Example 2

Every integer is a rational number.

Seven is a rational number.

Therefore seven is an integer.

Example 3

Every differentiable function is continuous.

The function f is not continuous.

Therefore

More about arguments in a few days! This is how we construct proofs!

To study logic efficiently we need a shorthand, consistent "language" to represent statements and ways to connect statements!

Sentential Logic - consists of an alphabet of symbols and rules to form sentences from the symbols.

Symbols

- ① specific basic sentence
- ② variable basic sentence
- ③ connectives
- ④ grouping

Definition: A sentence in sentential logic is

Rules

1. A specific basic sentence symbol or a variable basic sentence symbol is a sentence.
2. If **B** and **C** are sentences then so are
3. Only strings obtained by applying rules 1 and 2 a finite number of times are sentences.

Examples: Which of these strings of symbols are sentences?

① A

② $p \wedge q$

③ $(A \rightarrow B) \wedge A$

④ $A \wedge \sim B \wedge C$

⑤ $q \rightarrow (p \vee q)$

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Use the given dictionary to translate each English sentence into a formal sentential logic statement and each formal statement into an English sentence.

Dictionary:

P: The number n is prime.

Q: The number n is rational.

S: The number n is the square root of an integer.

Z: The number n is an integer.

1. English to Formal

- (a) The number n is a prime integer.
- (b) The number n is rational exactly when n is the square root of an integer.
- (c) If the number n is the square root of an integer but n is also an integer, then n is prime.
- (d) The number n is rational when n is the square root of an integer.
- (e) Either the number n is prime and n is an integer, or n is rational.

2. Formal to English

- (a) $P \vee Q$
- (b) $P \rightarrow [(\sim P) \wedge S]$
- (c) $P \leftrightarrow (\sim Q)$
- (d) $(\sim P) \vee [Z \wedge (\sim Q)]$
- (e) $\sim(P \wedge Z)$

Sudoku Proofs: Clear Writing Practice

Rules for Sudoku

1. Row Rule: Each row must contain the numbers 1 to 9 exactly once.
2. Column Rule: Each column must contain the numbers 1 to 9 exactly once.
3. Box Rule: Each of the nine 3 x 3 subgrids must contain the numbers 1 to 9 exactly once.

Class Activity: Divide class into teams of 2 or 3 each. Starting with the Sudoku puzzle given below, fill in the five remaining 1's using the rules above. For each separate 1 added, carefully explain the reasoning used in terms of the rules. Devise a clear, consistent, concise notation to help with your explanations.

					4			6
		2		5	7			3
	4					1		
	2			1			4	
	3	9				6	5	
	1			3			8	
		7					6	
1			5	2		7		
6			4					

Puzzle consists of 9
boxes. Each box consists
of 9 cells.

Boxes and cells are labeled

The same as a phone keypad

1 2 3
4 5 6
7 8 9 •

What is the first number you can put in the puzzle below? Explain your reasoning.

	4	3		8		2	5	
6								
					1		9	4
9					4		7	
			6		8			
	1		2					3
8	2		5					
								5
	3	4		9		7	1	